

Videregående Mekanik Formelsamling

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1 Basics

1.1 Newton

$$\mathbf{F} = m\mathbf{a} = m\ddot{\mathbf{r}} = \dot{\mathbf{p}}$$

Hvis der ikke er en udefrapåvirkende kraft er impulsen bevaret.

1.2 Trigonometri

$$\begin{aligned}\sin(\theta \pm \phi) &= \sin \theta \cos \phi \pm \cos \theta \sin \phi \\ \cos(\theta \pm \phi) &= \cos \theta \cos \phi \mp \sin \theta \sin \phi \\ \cos \theta \cos \phi &= \frac{1}{2} [\cos(\theta + \phi) + \cos(\theta - \phi)] \\ \sin \theta \sin \phi &= \frac{1}{2} [\cos(\theta - \phi) - \cos(\theta + \phi)] \\ \sin \theta \cos \phi &= \frac{1}{2} [\sin(\theta + \phi) + \sin(\theta - \phi)] \\ \cos^2 \theta &= \frac{1}{2} [1 + \cos 2\theta] \\ \sin^2 \theta &= \frac{1}{2} [1 - \cos 2\theta] \\ \cos \theta + \cos \phi &= 2 \cos \left(\frac{\theta + \phi}{2} \right) \cos \left(\frac{\theta - \phi}{2} \right) \\ \cos \theta - \cos \phi &= -2 \sin \left(\frac{\theta + \phi}{2} \right) \sin \left(\frac{\theta - \phi}{2} \right) \\ \sin \theta + \sin \phi &= 2 \sin \left(\frac{\theta + \phi}{2} \right) \cos \left(\frac{\theta - \phi}{2} \right) \\ \sin \theta - \sin \phi &= 2 \cos \left(\frac{\theta + \phi}{2} \right) \sin \left(\frac{\theta - \phi}{2} \right) \\ \cos^2 \theta + \sin^2 \theta &= 1 \\ \sec^2 \theta - \tan^2 \theta &= 1 \\ e^{i\theta} &= \cos \theta + i \sin \theta \quad [\text{Euler's relation}] \\ \cos \theta &= \frac{1}{2} (e^{i\theta} + e^{-i\theta}) \\ \sin \theta &= \frac{1}{2i} (e^{i\theta} - e^{-i\theta})\end{aligned}$$

1.3 Series Expansions

$$f(z) = f(a) + f'(a)(z-a) + \frac{1}{2!}f''(a)(z-a)^2 + \frac{1}{3!}f'''(a)(z-a)^3 + \dots \quad [\text{Taylor's series}]$$

$$e^z = 1 + z + \frac{1}{2!}z^2 + \frac{1}{3!}z^3 + \dots$$

$$\cos z = 1 - \frac{1}{2!}z^2 + \frac{1}{4!}z^4 - \dots$$

$$\cosh z = 1 + \frac{1}{2!}z^2 + \frac{1}{4!}z^4 + \dots$$

$$\tan z = z + \frac{1}{3}z^3 + \frac{2}{15}z^5 + \dots \quad [|z| < \pi/2]$$

$$(1+z)^n = 1 + nz + \frac{n(n-1)}{2!}z^2 + \dots \quad [|z| < 1] \quad [\text{binomial series}]$$

$$\ln(1+z) = z - \frac{1}{2}z^2 + \frac{1}{3}z^3 - \dots \quad [|z| < 1]$$

$$\sin z = z - \frac{1}{3!}z^3 + \frac{1}{5!}z^5 - \dots$$

$$\sinh z = z + \frac{1}{3!}z^3 + \frac{1}{5!}z^5 + \dots$$

$$\tanh z = z - \frac{1}{3}z^3 + \frac{2}{15}z^5 + \dots \quad [|z| < \pi/2]$$

1.4 Integrals

$$\int \frac{dx}{1+x^2} = \tan^{-1} x$$

$$\int \frac{dx}{1-x^2} = \tanh^{-1} x$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x$$

$$\int \frac{dx}{\sqrt{1+x^2}} = \sinh^{-1} x$$

$$\int \tan x \, dx = -\ln \cos x$$

$$\int \tanh x \, dx = \ln \cosh x$$

$$\int \frac{dx}{x+x^2} = \ln \left(\frac{x}{1+x} \right)$$

$$\int \frac{x \, dx}{1+x^2} = \ln(1+x^2)$$

$$\int \frac{dx}{\sqrt{x^2-1}} = \cosh^{-1} x$$

$$\int x \, dx \sqrt{1+x^2} = \sqrt{1+x^2}$$

$$\int \frac{dx}{x\sqrt{x^2-1}} = \cos^{-1} \left(\frac{1}{x} \right)$$

$$\int \frac{\sqrt{x} \, dx}{\sqrt{1-x}} = \sin^{-1}(\sqrt{x}) - \sqrt{x(1-x)}$$

$$\int \frac{dx}{(1+x^2)^{3/2}} = \frac{x}{(1+x^2)^{1/2}}$$

$$\int \ln(x) \, dx = x \ln(x) - x$$